Version 3 complete classifier comparison — Original-only vs Multi-source

Internal: 49,229 frozen URLs | External: 10,000 identity-disjoint URLs | Cell values are percentages

A. Internal original test — detection metrics

	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
CNN-ORIG	98.9	97.5	97.3	97.4	99.7	99.3
CNN-MS	98.9	97.6	97.2	97.4	99.7	99.4
LSTM-ORIG	99.0	98.2	97.0	97.6	99.7	99.3
LSTM-MS	98.8	97.9	96.5	97.2	99.6	99.1
RNN-ORIG	98.7	97.3	96.3	96.8	99.4	98.8
RNN-MS	98.2	95.3	96.1	95.7	99.3	98.6
RF-ORIG	95.0	97.2	78.2	86.7	96.3	93.3
RF-MS	94.9	97.2	77.9	86.4	96.4	93.4
SVM-ORIG	89.7	96.9	52.8	68.4	93.0	87.2
SVM-MS	89.9	96.8	53.5	68.9	93.0	87.2
HYB-ORIG	99.1	98.8	96.9	97.8	99.8	99.5
HYB-MS*	96.1	100.0	81.3	89.7	99.6	99.2

B. Internal original test — error rates

	FPR	FNR
CNN-ORIG	0.67	2.7
CNN-MS	0.65	2.8
LSTM-ORIG	0.48	3.0
LSTM-MS	0.55	3.5
RNN-ORIG	0.71	3.7
RNN-MS	1.2	3.9
RF-ORIG	0.60	21.8
RF-MS	0.60	22.1
SVM-ORIG	0.45	47.2
SVM-MS	0.46	46.5
HYB-ORIG	0.31	3.1
HYB-MS*	0.01	18.7

C. Frozen external 10K — detection metrics

	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
CNN-ORIG	48.2	49.1	96.0	65.0	57.3	59.8
CNN-MS	48.9	49.4	96.6	65.4	79.5	80.7
LSTM-ORIG	47.9	48.9	94.3	64.4	83.2	86.7
LSTM-MS	51.6	50.9	96.4	66.6	93.1	95.4
RNN-ORIG	48.0	49.0	96.0	64.9	77.7	81.0
RNN-MS	48.3	49.1	96.1	65.0	87.3	89.3
RF-ORIG	32.9	38.6	58.0	46.3	24.8	42.6
RF-MS	48.3	48.8	67.6	56.7	63.0	74.3
SVM-ORIG	34.7	25.9	16.4	20.1	23.4	42.8
SVM-MS	34.6	26.7	17.6	21.2	25.0	43.7
HYB-ORIG	48.6	49.3	95.6	65.0	72.7	71.0
HYB-MS*	91.8	98.0	85.2	91.2	96.6	97.5

D. Frozen external 10K — error rates

	FPR	FNR
CNN-ORIG	99.6	4.0
CNN-MS	98.9	3.4
LSTM-ORIG	98.4	5.7
LSTM-MS	93.2	3.6
RNN-ORIG	99.9	4.0
RNN-MS	99.5	3.9
RF-ORIG	92.3	42.0
RF-MS	70.9	32.4
SVM-ORIG	47.0	83.6
SVM-MS	48.3	82.4
HYB-ORIG	98.5	4.4
HYB-MS*	1.7	14.8

ORIG = 358,619 original training URLs; MS = 363,692 multi-source real URLs; no GAN rows. Baseline and HYB-ORIG cells are means across seeds 42/43/44. *HYB-MS is one calibrated three-seed ensemble result. Higher is better for A/C; lower is better for FPR/FNR in B/D.